

SAYAM SETHI

Email  | Google Scholar  | GitHub  | LinkedIn 

EDUCATION

PhD in Electrical and Computer Engineering

August 2023 – 2028 (tentative)

Advisor: Dr. Jonathan Baker; The University of Texas at Austin

Austin, Texas

GPA: 3.90/4.00

Bachelor of Technology | Dept. of Computer Science and Engineering

July 2019 – May 2023

Indian Institute of Technology, Delhi

Delhi, India

GPA: 9.55/10.00

PUBLICATIONS¹

Optimizing Parallel Execution of Commuting Pauli Product Rotations

May 2026

S. Sethi, D. Nambisan, and J.M. Baker

arXiv

Price and Payoff: Non-Determinism in Fault Tolerant Quantum Computation

May 2026

A. Awasthi, S. Sethi, S. Khan, G.S. Ravi, and J.M. Baker

QCE '26

INJEQT: Improved Magic-State Injection Protocol for Fault-Tolerant Quantum Extractor Architectures

April 2026

S. Sethi, S. Khan, A. Awasthi, A. Anand, and J.M. Baker

QCE '26

Architecting Early Fault Tolerant Neutral Atoms Systems with Quantum Advantage

April 2026

S. Khan, S. Sethi, K. Sahay, Y. Lin, J. Alnas, S. Kurapati, A. Anand, J.M. Baker, and K.R. Brown

Under review

Logical Compilation for Multi-Qubit Iceberg Patches

April 2026

C. Mazzetti, S. Sethi, R. Rines, P. Gokhale, and J.M. Baker

Under review

Optimizing Logical Mappings for Quantum Low-Density Parity Check Codes

March 2026

S. Sethi, S. Khan*, M. Poster, A. Anand, and J.M. Baker*

Under review

RESCQ: Realtime Scheduling for Continuous Angle QEC Architectures

March 2025

S. Sethi, and J.M. Baker

ASPLOS '25

Cyclic Qubit Mappings

September 2024

M. Poster, S. Sethi, and J.M. Baker

QCE '24

3D-TemPo: Optimizing 3-D DRAM Performance Under Temperature and Power Constraints

February 2024

S. Pandey, S. Sethi, and P.R. Panda

IEEE TCAD

INTERNSHIPS

Infleqtion

May 2026 - August 2026; 7 September 2026 - January 2027

Graduate Quantum Software Intern

IBM Research, TJ Watson Research Center

Summer 2024 and Summer 2025

Quantum Research Scientist Intern

Quadeye Securities LLP, Gurugram

Jun – Jul 2022

Systems Intern

¹ * denotes equal contribution

INVITED TALKS AND POSTERS

Exploring Asymmetric QEC Code Concatenation (poster)

8th International Quantum Error Correction conference

June 2026

Santa Barbara, CA

INJEQT: Improved Magic-State Injection Protocol for Fault-Tolerant Quantum Extractor Architectures

May 2026

IBM Circuit and Compilers Group

Virtual

RESCQ: Realtime Scheduling for Continuous Angle QEC Architectures

IBM Circuit and Compilers Group

UT Austin CompArch Seminar

April, May 2025

Yorktown Heights, NY

Austin, TX

PROFESSIONAL SERVICES

- Artifact Evaluation PC for 59th IEEE/ACM International Symposium on Microarchitecture (MICRO 2026).
- Junior PC for Symposium on Parallelism in Algorithms and Architectures (SPAA 2025).

ACADEMIC ACHIEVEMENTS

- **Texas Quantum Institute Graduate Fellowship (2025-27):** Awarded a fellowship for investigating classical overheads in FTQC compilation
- **IIT Delhi Semester Merit Award:** Secured for being in the **top 7 percentile** in **Semester I** of **2019-20** and **2021-22**
- **ICPC India Regionals, Amritapuri '21:** Secured **India Rank 47** and **Institute Rank 1** out of **4393 teams**
- **Google HashCode '22:** Secured **India Rank 12**, **Institute Rank 1** and **Global Rank 240** out of **10177 teams**
- **Indian National Mathematical Olympiad:** Qualified for **INMO 2018** (precursor to IMO) on being shortlisted in the Telangana region in **RMO 2017**
- **Kishore Vaigyanik Protsahan Yojana (KVPY):** Offered the fellowship in **SX stream (2018-19)** and **SA stream (2017-18)** with **All India Ranks** of **34** and **271** respectively

TEACHING EXPERIENCE AND EXTRA-CURRICULAR ACTIVITIES

Teaching Assistant

- Grad TA for ECE 412 - Software Design & Implementation in Fall '26, course for ECE sophomores
- Grad TA for ECE 382V - Quantum Error Correction Theory & Systems in Spring '26
- Grad TA for ECE 312 - Software Design & Implementation in Fall '24, Spring '25, course for ECE sophomores
- Grad TA for ECE 306 - Introduction to Computing in Fall '23, introductory course for ECE freshers
- Head TA for COL216 (**received the outstanding TA award**) - Computer Architecture in Spring '23, mandatory course for CSE sophomores
- TA for COL215 - Digital Logic and System Design in Fall '22, mandatory course for CSE sophomores

Algorithms and Coding Club, IIT Delhi

- **Overall Coordinator, 2022-23:** Elevated the club's recognition from a Chapter to a Technical Club; organized events in Tryst, Technical Fest at IITD
- **Coordinator, 2021-22:** Conducted the Summer of CP and took live lectures on YouTube; guided executives
- **Executive, 2020-21:** Authored various problems for Competitive Programming contests conducted by the club

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, VHDL, Assembly, Standard ML, Octave, MATLAB, R

Libraries/Frameworks and Tools: Qiskit, Stim, Sniper, HotSpot, CoMeT, Vivado, L^AT_EX, git